

● MEDIUM

● ADVANCED MATH

# Nonlinear functions

10 questions · ~15 min

06

**Q1**

ADVANCED MATH

The function  $f$  is defined by  $fx = 7x^3$ . In the  $xy$ -plane, the graph of  $y = gx$  is the result of shifting the graph of  $y = fx$  down 2 units. Which equation defines function  $g$ ?

A.  $gx = \frac{7}{2}x^3$

B.  $gx = 7x^{\frac{3}{2}}$

C.  $gx = 7x^3 + 2$

D.  $gx = 7x^3 - 2$

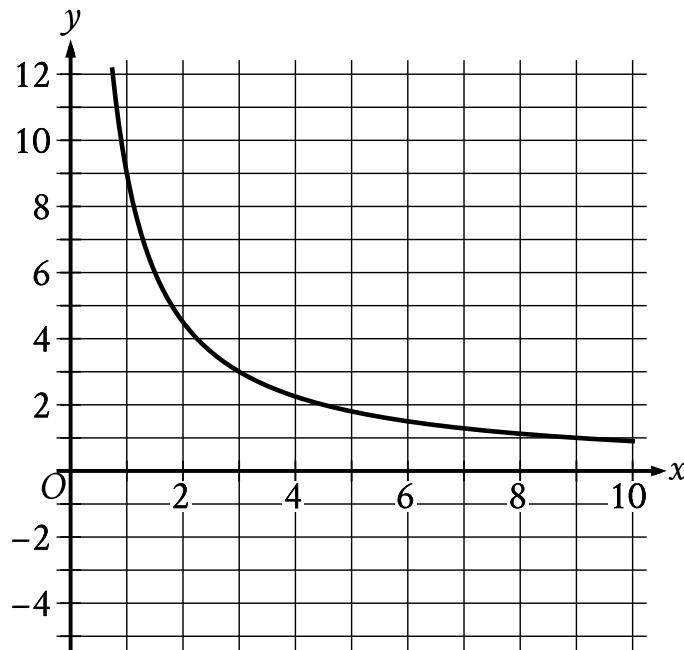
From 2005 through 2014, the number of music CDs sold in the United States declined each year by approximately 15% of the number sold the preceding year. In 2005, approximately 600 million CDs were sold in the United States. Of the following, which best models  $C$ , the number of millions of CDs sold in the United States,  $t$  years after 2005?

A.  $C = 600(0.15)^t$

B.  $C = 600(0.85)^t$

C.  $C = 600(1.15)^t$

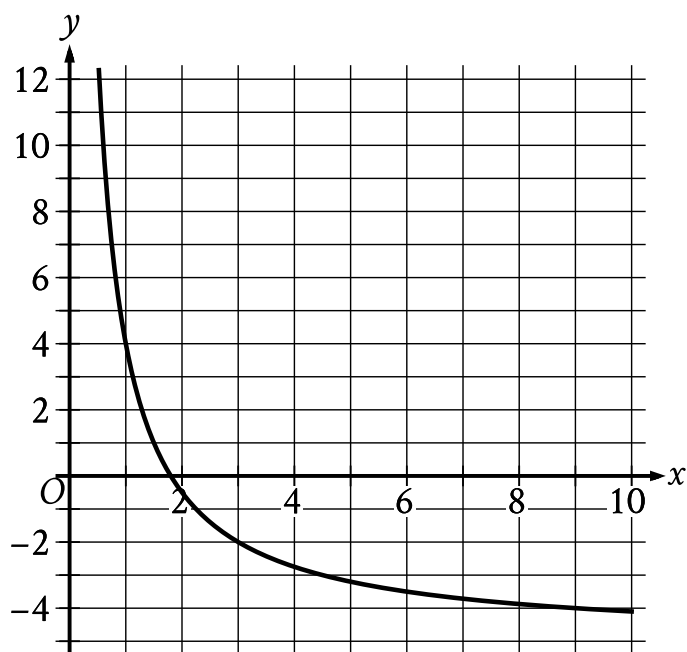
D.  $C = 600(1.85)^t$



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to  $x$  equals 3.
  - The curve then trends down gradually.
- As  $x$  increases, the curve approaches the line  $y$  equals 0.
- The curve passes through the following points:
  - (2 comma nine halves)
  - (3 comma 3)
  - (6 comma three halves)

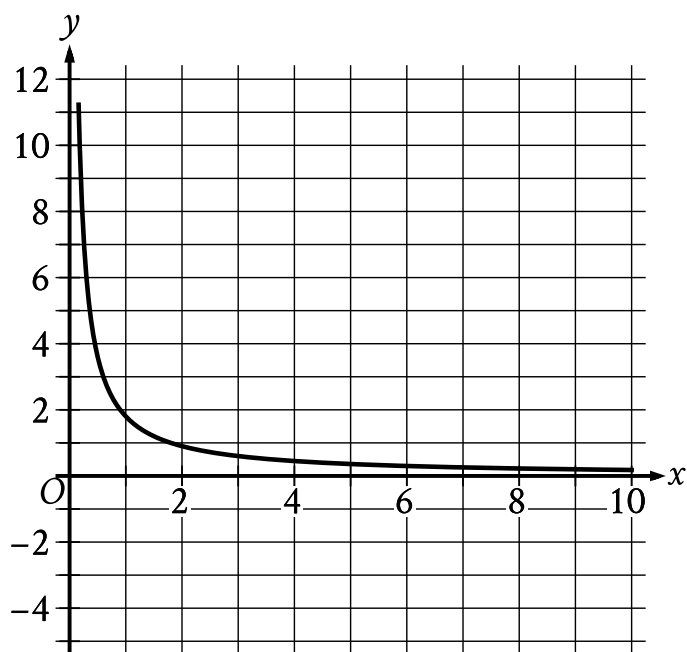
The graph of the rational function  $f$  is shown, where  $y = f(x)$  and  $x \geq 0$ . Which of the following is the graph of  $y = f(x) + 5$ , where  $x \geq 0$ ?

A.



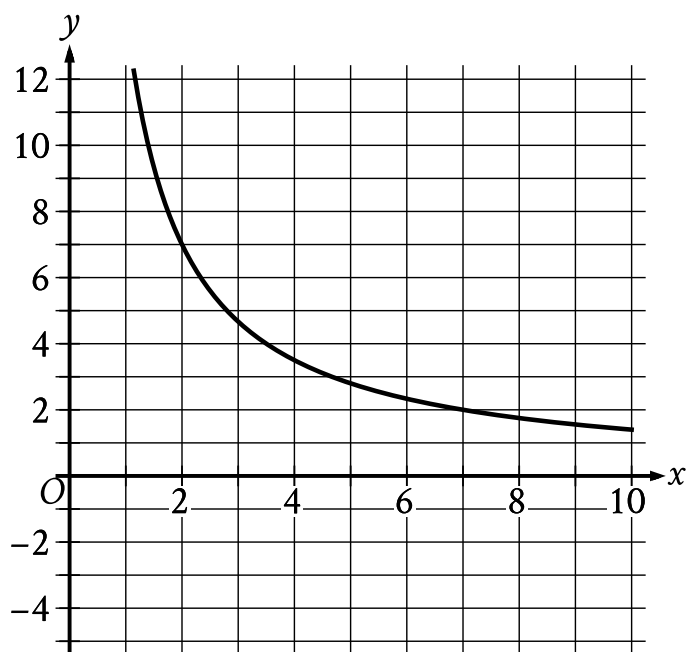
- Moving from left to right:
  - The curve passes from quadrant 1 to quadrant 4.
  - The curve trends down sharply to  $x$  equals 3.
  - The curve then trends down gradually.
- As  $x$  increases, the curve approaches the line  $y$  equals negative 5.
- The curve passes through the following points:
  - (2 comma negative one half)
  - (3 comma negative 2)
  - (6 comma negative seven halves)

B.



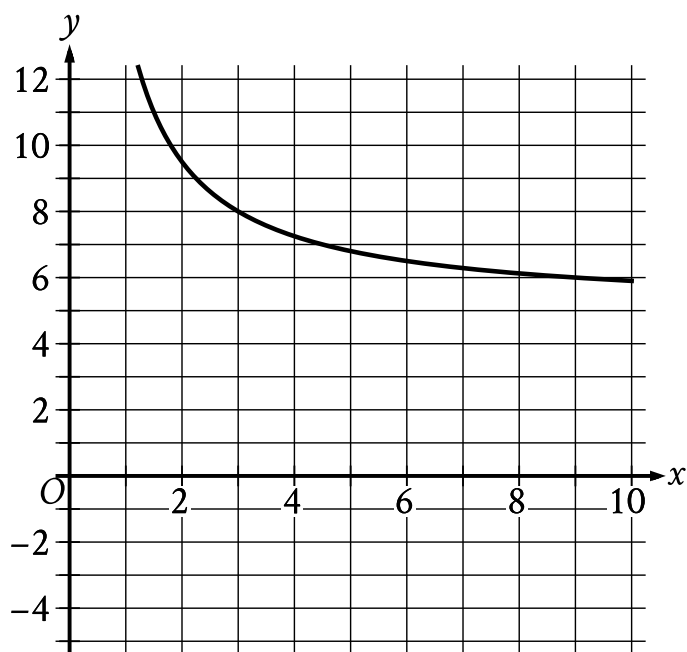
- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to  $x$  equals 1.
  - The curve then trends down gradually.
- As  $x$  increases, the curve approaches the line  $y$  equals 0.
- The curve passes through the following points:
  - (2 comma nine tenths)
  - (3 comma three fifths)
  - (6 comma three tenths)

C.



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to  $x$  equals 2.
  - The curve then trends down gradually.
- As  $x$  increases, the curve approaches the line  $y$  equals 0.
- The curve passes through the following points:
  - (2 comma 7)
  - (3 comma  $\frac{14}{3}$ )
  - (6 comma  $\frac{7}{3}$ )

D.



- Moving from left to right:
  - The curve is shown in quadrant 1.
  - The curve trends down sharply to  $x$  equals 3.
  - The curve then trends down gradually.
- As  $x$  increases, the curve approaches the line  $y$  equals 5.
- The curve passes through the following points:
  - $(2, \frac{19}{2})$
  - $(3, 8)$
  - $(6, \frac{13}{2})$



The function  $f(t) = 40,000 \cdot 2^{\frac{t}{790}}$  gives the number of bacteria in a population  $t$  minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

- A. 2
- B. 790
- C. 1,580
- D. 40,000

|      |    |    |    |
|------|----|----|----|
| $x$  | 24 | 30 | 32 |
| $fx$ | -8 | -8 | 8  |

For the quadratic function  $f$ , the table shows three values of  $x$  and their corresponding values of  $fx$ . Which equation defines  $f$ ?

- A.  $f(x) = (x - 24)(x - 30) + 4$
- B.  $f(x) = (x - 24)(x - 30) - 8$
- C.  $f(x) = (x - 8)(x - 32) + 32$
- D.  $f(x) = (x - 8)(x - 32) - 32$

The equation  $E(t) = 51.8^t$  gives the estimated number of employees at a restaurant, where  $t$  is the number of years since the restaurant opened. Which of the following is the best interpretation of the number 5 in this context?

- A. The estimated number of employees when the restaurant opened
- B. The increase in the estimated number of employees each year
- C. The number of years the restaurant has been open
- D. The percent increase in the estimated number of employees each year

Immanuel purchased a certain rare coin on January 1. The function  $f(x) = 651.03^x$ , where  $0 \leq x \leq 10$ , gives the predicted value, in dollars, of the rare coin  $x$  years after Immanuel purchased it. What is the best interpretation of the statement “ $f(8)$  is approximately equal to 82” in this context?

- A. When the rare coin's predicted value is approximately 82 dollars, it is 8% greater than the predicted value, in dollars, on January 1 of the previous year.
- B. When the rare coin's predicted value is approximately 82 dollars, it is 8 times the predicted value, in dollars, on January 1 of the previous year.
- C. From the day Immanuel purchased the rare coin to 8 years after Immanuel purchased the coin, its predicted value increased by a total of approximately 82 dollars.
- D. 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.

A rectangle has a length that is 15 times its width. The function  $y = 15ww$  represents this situation, where  $y$  is the area, in square feet, of the rectangle and  $y > 0$ . Which of the following is the best interpretation of  $15w$  in this context?

- A. The length of the rectangle, in feet
- B. The area of the rectangle, in square feet
- C. The difference between the length and the width of the rectangle, in feet
- D. The width of the rectangle, in feet

A model predicts that the population of Bergen was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population  $p$  increased by 4% of the previous year's population. Which equation best represents this model, where  $x$  is the number of years after 2005, for  $x \leq 5$ ?

A.  $p = 0.9615,000^x$

B.  $p = 1.0415,000^x$

C.  $p = 15,0000.96^x$

D.  $p = 15,0001.04^x$

An egg is thrown from a rooftop. The equation  $h = -4.9t^2 + 9t + 18$  represents this situation, where  $h$  is the height of the egg above the ground, in meters,  $t$  seconds after it is thrown. According to the equation, what is the height, in meters, from which the egg was thrown?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.